

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			Indicative content: Names: The electromagnetic spectrum includes radio waves, microwaves, infra-red, visible light, ultraviolet, X-rays and gamma rays. Similarities: They are all transverse waves. All regions of the electromagnetic spectrum transfer energy and also transmit information. They all travel at the same speed in space. Differences: Gamma rays have the shortest wavelength and highest frequency/energy. They have different uses (some maybe stated) and different ionising properties. 5-6 marks All parts named, similarities and differences correctly identified. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3-4 marks Addresses 2 areas well out of parts named, similarities and differences correctly identified OR limited description provided of each area. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i>	6			6		

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
				1-2 marks Addresses 1 area well out of parts named, similarities and differences correctly identified OR limited description provided of up to 2 areas. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
				Question 5 total	6	0	0	6	0	0